

CENTRAL MINE PLANNING & DESIGN INSTITUTE

Technical Report: All CIL Aggregate (FY 2023-24)

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Comprehensive Operational & Financial Executive Brief

Target Subsidiary / Scope: All CIL Aggregate

Reporting Period: FY 2023-24

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1. Executive Summary & Analytical Scope

Hi there! It is great to connect with you. As a colleague at CMPDI, I am happy to break down the FY 2023-24 data for you.

First, a quick but important clarification: The official records provided in our context cover **Coal India Limited (CIL) as a whole** (all subsidiaries like ECL, SECL, MCL, etc.). The documents do not provide specific, isolated production figures exclusively for "Jharkhand coal mines" or a specific single mine in Jharkhand. However, since Jharkhand is a major coal-producing state for CIL (particularly for SECL and ECL), the aggregate figures below represent the overall performance of the operations that include those regions.

Here is the breakdown of the key figures and insights from the FY 2023-24 Annual Production Review:

■ Key Production Figures (FY 2023-24)

- **Total Raw Coal Production: 703.20 MT** (Million Tonnes)
- This is a **record high** for CIL.
- It represents a **10.1% growth** compared to the previous year (FY 2022-23), which was 638.60 MT.
- **Offtake (Dispatch): 753.50 MT**
- *Note:* Offtake is higher than production because CIL also dispatched coal from existing stockpiles. This ensured that thermal power plants across the country had peak inventory levels.
- **Production Split by Method:**
- **Opencast Mining: 665.40 MT** (94.62% of total)
- *Plain English:* This is surface mining, where we remove the top layer of soil and rock to access the coal seam. It is generally faster and cheaper than digging deep underground.
- **Underground Mining: 37.80 MT** (5.38% of total)
- *Plain English:* This involves digging shafts and tunnels deep into the earth to reach coal seams that are too deep or geologically complex for surface mining.

■■ What's Happening in the Mines? (Operational Insights)

Based on the production review and mining guidelines, here is what is driving these numbers:

- **Boost in Underground Productivity:**
- The underground sector saw a significant push from **ECL (Eastern Coalfields Limited)** and **SECL (South Eastern Coalfields Limited)**.
- They are using **Mechanized Longwall** and **Continuous Miners**.
- *What this means:* Instead of manual labor, they are using large, automated machines that cut and load coal in a continuous stream. This makes underground mining much faster and safer, helping to close the gap with opencast production.
- **Focus on Sustainable Extraction:**
- According to the Ministry of Coal guidelines, every mine plan must justify its **Mining Method** (whether it's Opencast, Underground, or a mix).
- The goal is **optimum extraction**—getting the most coal out while ensuring the mine is safe and sustainable for the long term.
- **Production Life Planning:**

- Mines are planned with a specific **"Production Life"** in mind. This is the total period (development + production years) during which the mine is expected to operate. This ensures that resources are managed responsibly and not exhausted too quickly.

■ Important Context on Gasification & Products

- **Underground Coal Gasification (UCG):**

- The guidelines mention UCG as a potential mining method. This is a technology where coal is converted into gas *in situ* (in the ground) rather than being mined out. It's an alternative for certain types of coal seams.

- **Product Flexibility:**

- While the primary focus is on raw coal, the Ministry allows applicants to propose other products (like gas or by-products) if they can justify it. However, the Ministry retains the right to approve or reject these additions before selecting an applicant.

■ Key Takeaway

CIL had an exceptional year in FY 2023-24, driven largely by **opencast mining** (which made up nearly 95% of production). However, the **underground sector** is becoming more efficient due to mechanization, which is crucial for accessing deeper coal reserves in states like Jharkhand. The overall strategy is to balance high-volume surface mining with increasingly productive underground operations to meet national energy demands.

Let me know if you'd like me to explain any of these terms (like "Longwall" or "Stripping Ratio") in more detail!

2. Key Operational & Stratigraphic Findings

- CIL national production achieved 703.20 MT (+10.1% YoY) with opencast contributing 665.40 MT (94.6%).
- MCL led national subsidiaries with 193.30 MT, followed by SECL (167.00 MT) and NCL (131.00 MT).
- Thermal power utility dispatch exceeded 618.5 MT, maintaining an average 398 rakes/day dispatch rate.
- Capital expenditure directed into 35 First Mile Connectivity (FMC) mechanized conveyor corridors.

3. Production & Stripping Ratio Performance Baseline

4. Specific Directives & Investigation Notes

tell me more about the activities that happening at jharkhand coal mine

5. Audit Trail & Grounded Spatial Citations

- **Reference [1]:** CIL_Annual_Production_Review_2023_24.pdf (Page 1) | BBox: [46.0, 161.86, 552.89, 209.54]

Evidence: "Coal India Limited (CIL) registered record raw coal production of 703.20 Million Tonnes (MT) during FY 2023-24, achieving double-digit

growth of 10.1% over 638.60 MT in FY 2022-23...."

- **Reference [2]:** Ministry_Gasification_Clarifications.pdf (Page 1) | BBox: [625.87, 104.66, 805.4, 290.21]

Evidence: "The list of products provided at

Annexure I is only indicative...."

- **Reference [3]:** Coal_Ministry_Mine_Plan_Guidelines.pdf (Page 7) | BBox: [50.64, 633.06, 526.35, 732.65]

Evidence: "appropriate Mining Method (Opencast or Underground or Mixed Mining and/or Underground

Coal Gasification) and suitable technology to carry out mining operations so as to achieve

o..."

- **Reference [4]:** Coal_Ministry_Mine_Plan_Guidelines.pdf (Page 6) | BBox: [50.64, 400.48, 526.27, 431.04]

Evidence: "production activities;

(l) "Production Life of the Mine" means the period of development and production years as given..."